OPERATING & MAINTENANCE

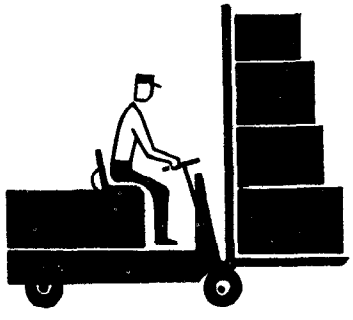
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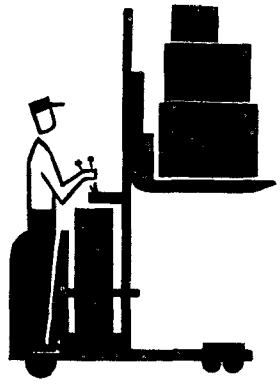
INSTRUCTIONS

C-185—Model 201
C-290—Model 301

IC 4483

SCR CONTROL for ELECTRIC VEHICLES





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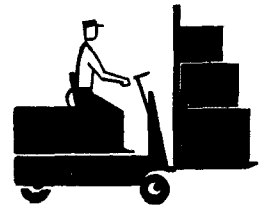
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The information contained herein is intended to assist truck users and dealers in the servicing of SCR control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance.

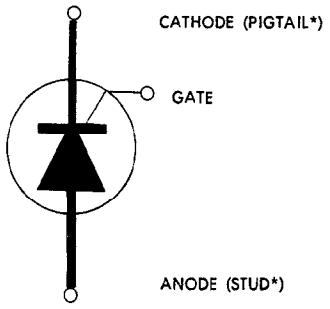
Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the truck manufacturer through his normal service channels, not directly to General Electric Company.

GENERAL ELECTRIC

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WHAT IS AN SCR?

Since the heart of the control is a silicon controlled rectifier (SCR), a general understanding of the characteristics of the device will be helpful. The SCR is a semi-conductor rectifier used as a latching switch; i.e., it may assume either a conducting or nonconducting state (On or Off).



The SCR can be turned on by a momentary application of control current to the gate. To turn it off, it is necessary in addition to removing the turn-on signal from the gate, either to remove all power from the SCR or to apply momentary reverse voltage between cathode and anode.

*** Typical of SCR as used in GE control for electric vehicles.**

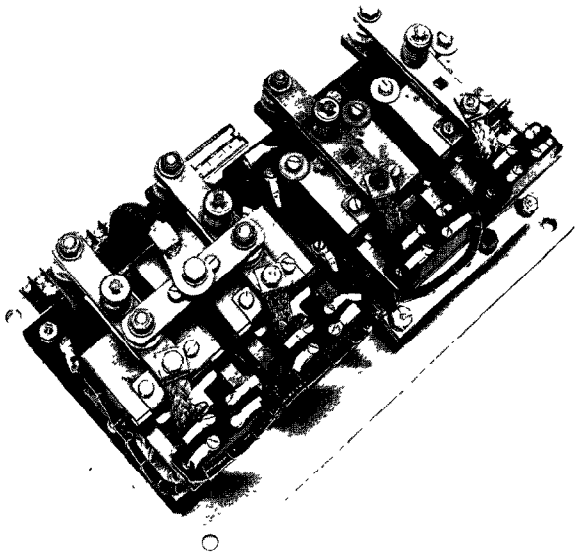


Figure 2
Typical magnetic panel consisting of forward, reverse, bypass, and pump contactors.

PHOTOS OF CONTROL

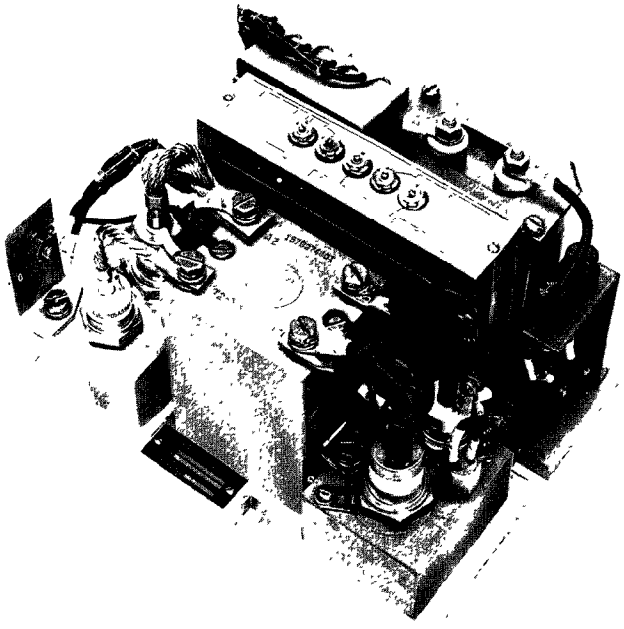


Figure 1
Typical SCR static panel. Model 201

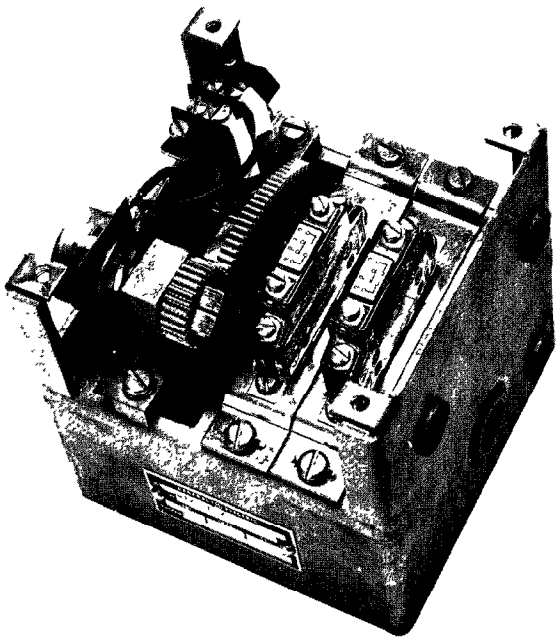
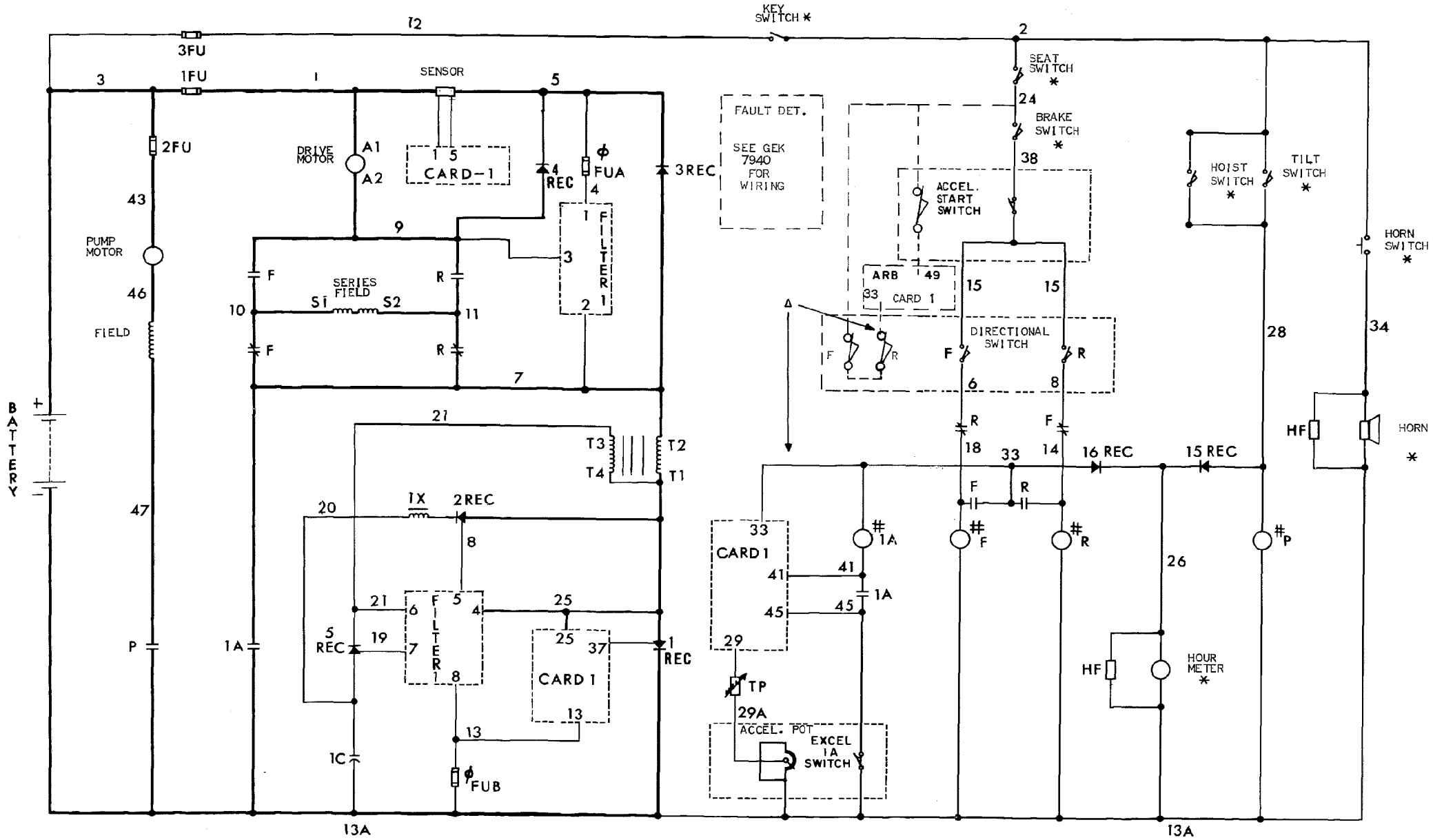


Figure 3
Typical foot-operated accelerator switch with the cover removed showing the speed control potentiometer and control interlocks.



This diagram is typical and may not necessarily reflect all the details of every truck application. Refer to the diagram furnished with the vehicle. Wire numbers may be preceded with a "G" to distinguish GE numbers from truck manufacturer's wire numbers.

Figure 4. Typical elementary diagram, General Electric C-185/290 SCR Control, Models 201 and 301

HF HASH FILTER
ϕ FUSES USED ON U/L APPROVED TRUCKS
COILS HAVE INTEGRAL FILTERS
Δ NOTE CHANGE OF WIRE NECESSARY TO CARD 1-33 FOR ANTI-ROLL-BACK OPTION (ARB)
* FURNISHED BY TRUCK MFG.

CIRCUIT OPERATION

(See Figure 4)

The circuit is energized by closing the key switch, the seat switch, the brake switch and moving the Forward or Reverse lever to either position and then depressing the accelerator closing the accelerator start switch. The F or R contactor coil is now energized, applying power to the drive motor circuit. Positive control power is fed through F or R interlock to wire 33, through the 1A coil to wire 41, to an oscillator located in Card 1.

The oscillator section will oscillate only when it receives both positive power through the F or R interlock, and a synchronizing control signal from the anode of 1 REC (wire 25). The oscillator output is fed from terminal 37 to the gate of 1 REC, the main SCR. This is the gate signal which will switch 1 REC to the conducting state. When 1 REC is conducting, current flow is from battery positive through 1 FU, drive motor, T2-T1, 1 REC and back to the battery negative. The initial rising d-c current through T2-T1 induces a voltage from T4 to T3, drives T3 below the battery negative, causes current to flow from filter 1 (wire 19) to the gate of 5 REC, turning 5 REC on. Current then flows from the transformer secondary T4 through 1 REC, 1C, 5 REC and back to T3, charging 1C (wire 20) negative until the transformer saturates, reducing this current flow to zero, and turning off 5 REC. The voltage of T3 then swings from negative to positive, causes the current to flow from filter 1 (wire 8) to the gate to 2 REC, turning 2 REC on. 2 REC conducts, and capacitor 1C discharges around the circuit composed of 1C, 1 REC, 2 REC and 1X. This discharge current opposes the battery current through 1 REC so that the resultant current is zero. With reverse voltage across 1 REC (the main SCR), 1 REC is turned off.

This explanation has been for one complete cycle, or pulse, of circuit operation. Figure 5 illustrates the pulsing of current from the battery.

During off time, the energy stored in the motor, by virtue of its inductance, will cause current to circulate through the motor around the loop formed by 3 REC, thus providing what is called "flyback current." Figure 6 shows the nature of the motor current which is composed of both battery current and the inductive flyback current. It should be noted that the average motor current measured will be greater than the average battery current. The SCR control, in effect, converts battery current at battery volts into a higher motor current and a lower motor voltage.

The time for the start of the next cycle is determined by the time that the oscillator section of the card takes to oscillate. This frequency of oscillation is controlled by the potentiometer in the accelerator. Slow speed is obtained by having maximum ohms in the potentiometer. As the resistance in the pot decreases, the speed of the motor increases. With level operation, the SCR circuit is capable of delivering approximately 70-90% speed. For full-speed operation, the 1A contactor is closed to apply full battery voltage to the motor. The 1A coil is energized by closing the 1A switchette in the accelerator.

CARD 1

- CURRENT LIMIT – The current-limit section of Card 1 provides protection to the motor and control by limiting currents during acceleration and stall. This circuit is sensitive to load current and overrides the oscillator under heavy loads so as to limit the pulse frequency (thus the average current) to a value based on the maximum rating of 1 REC. Because of the flyback current through 3 REC, the motor current usually runs 2 to 3 times this current-limit value. The CURRENT LIMIT is adjustable by means of a trimpot on Card 1.
- OSCILLATOR – The oscillator section of the card has two adjustable modes and one fixed feature. With the accelerator pot at maximum resistance, the CREEP SPEED can be adjusted with a trimpot on the card. With the accelerator pot at minimum resistance, the TOP SCR SPEED is adjustable by means of a trimpot on the card. The fixed feature is controlled acceleration. When the accelerator is set for maximum speed and the directional switch is closed, the controlled acceleration provides a gradual buildup of pulses, thus giving a smooth acceleration to top SCR speed. This feature also provides a smooth re-acceleration during a plugging reversal of direction.
- PLUGGING – Slowdown is accomplished when reversing by providing a small amount of retarding torque for deceleration. If the truck is moving and the direction lever is moved from forward to reverse, the motor field is reversed. During the 1 REC off time the motor armature, driven by the inertia of the truck, acts as a generator. This generated current passes through 4 REC and a current sensor. A signal taken from the sensor is fed to Card 1 retarding the pulse frequency and provides a soft reverse stopping action. The distance or severity of the reversal is adjustable by means of a PLUGGING trimpot on the card.
- 1A TIMER – A time-delay pickup of 1A is provided by a circuit in Card 1. This allows the truck to accelerate through the SCR range before 1A picks up, even if the accelerator 1A switch is closed immediately. This time delay is adjustable by means of a 1A TIME trimpot on Card 1. The timer circuit is rendered inoperative any time that plugging is in process, or if the main SCR fails to pulse.
- RAMP RESTART OPTION – A portion of Card 1 can be connected through normally closed switches in the directional and start switches (see Fig. 4). Normally, when the truck is stopped on a ramp and the foot brake is used to hold it in position the truck will roll back when the operator moves his foot back to the accelerator. The ramp restart circuit prevents the control from assuming a plugging condition and allows for full torque up the ramp.

1A CONTACTOR (By-pass contactor around the SCR control) – The 1A contactor is used to provide top truck speed, torque, and efficiency when called for. The 1A contactor is picked up when the accelerator is moved to its extreme end of travel.

THERMAL PROTECTOR – A thermal protector (TP) is mounted on the heat sink between 1 REC and 2 REC. This is a temperature sensitive device which increases resistance with an increase in temperature. During the normal operating range, the thermal protector has a resistance of approximately 50 ohms. If the temperature of the 1 REC heat sink exceeds 80°C., the resistance of the thermal protector increases. Being in series with the accelerator potentiometer, this increased resistance decreases the speed of the truck. The truck will operate at a reduced speed until the temperature reaches a safe value, then full SCR power will be available.

Other functions and equipment available with the SCR control for electric vehicles and their instruction references are:

IC4484 FAULT DETECTOR . . . . . .(GEK-7940)
IC4484Y100D STATIC RETURN-TO-OFF CARD . . . . . .(GEK-7941)
IC4484 FIELD-WEAKENING SYSTEM . . . . . .(GEK-8475)
IC3012BH ACCELERATOR MASTER SWITCH . . . . . .(GEK-8073)

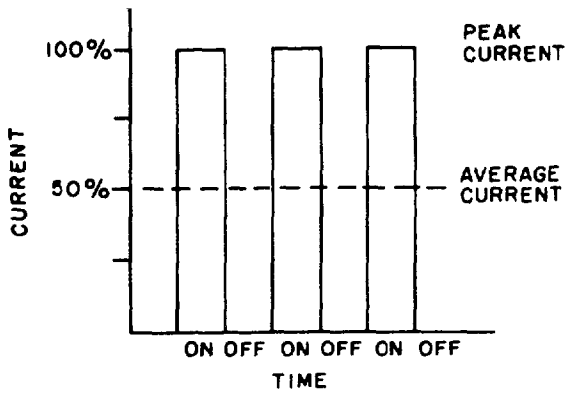


Figure 5. Battery current

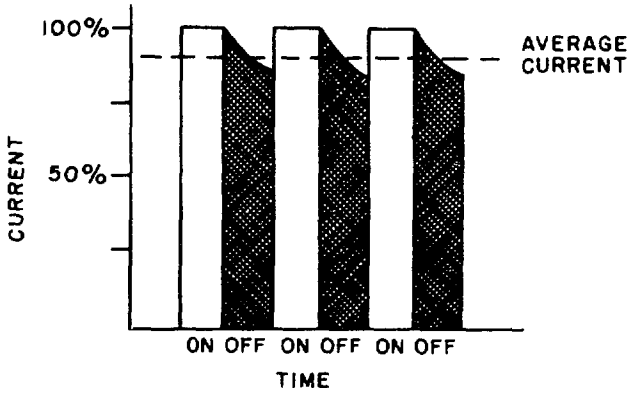


Figure 6. Motor current

General Maintenance Instructions

The SCR control, like all electrical apparatus, does have some thermal losses. The semiconductor junctions have finite temperature limits above which these devices may be damaged. For these reasons, normal maintenance should guard against any action which will expose the components to excessive heat, such as steam cleaning; or which will reduce the heat dissipating ability of the control, such as restricting air flow.

The following DO'S and DON'TS should be observed:

- Any controls that will be used in altitudes of 5000 feet or over and in ambients of 100°F (40°C) or over should be brought to the attention of the truck manufacturer.

- The control should not be steam cleaned. In dusty areas, use low pressure air to blow off the control. In oily or greasy areas, a mild solution of detergent or de-natured alcohol can be used to wash off the control. Then blow completely dry with low pressure air.

- For the SCR panel to be most effective, it must be mounted against the frame of the truck. The truck frame, acting as an additional heat-sink, will give improved truck performance by keeping the SCR Control package cooler.

- Terminal boards and other exposed SCR control parts should be kept free of dirt and paint which might change the effective resistance between points.

- The truck should not be plugged when the truck is jacked up and the drive wheels are in a free wheeling position. This can create excessive voltages that can be harmful to the control.

Trouble-Shooting Instructions

The pulsing of the main SCR is too fast for conventional instruments to measure. When the control is functioning properly, a low hum can be heard.

Malfunctions of the SCR will generally fall into one of two categories. These are either no power (Table 1) or full power (Table 2), when operating in the SCR control range.

These simple and easy-to-follow tables outline the various symptoms and the corrective action to be taken.

The same device designations have been maintained on different controls but the wire numbers may vary. Refer to the elementary and wiring diagram for your specific control. The wire numbers shown on the elementary diagram will have identical numbers on the corresponding wiring diagrams for a specific truck, but these numbers may be different from the numbers referenced in this publication. All wire numbers will be preceded with a "G" to distinguish GE numbers from truck manufacturer's wires.

Before proceeding, visually check for loose wiring, maladjusted linkage to accelerator switch, signs of overheating of components, etc. Before touching electrical components, disconnect the battery and discharge capacitor 1C. Reconnect the battery as needed for the specific check.

Tools and test equipment required are 36-volt test battery, 180-ohm 2-watt resistor, 3-volt battery – 3-volt lamp (or BRIGHT STAR No. 1618CT circuit continuity tester), clip leads, volt-ohmmeter (20,000 ohms per volt) and general hand tools.

Table 1

Failures Which Cause No Motor Torque With SCR Control

SYMPTOMS	WHAT TO DO
1A. The contactors do not pick up. No control voltage is present from positive to negative.	• Check the power fuses. • Check the battery for low specific gravity and the connections for looseness or broken fittings.
1B. The contactors do not pick up. Control volts are present from positive to negative.	• (For these tests, if a fault detector is used, disconnect wire 33 from fault detector terminal 3.) • (See NOTE 1) Connect a jumper from battery positive to the positive side of the F or the R coil. If the device does not pick up, check the coil for continuity. Also jumper negative to opposite terminal to check for opens in negative connections. • (See NOTE 1) With the jumper on battery positive move the other end of the jumper to wire 8 on the F interlock, or 6 on the R interlock. The coils should pick up. This proves F and R electrically interlock. • (See NOTE 1) Using the jumper continue to check the remaining components in the circuit such as the directional switch, brake switch, seat switch and key switch, by moving the end of the jumper to the positive side of each of these devices.
1C. The contactors close. No power and no SCR hum with the accelerator in the SCR range.	• (See NOTE 1) With F or R picked up and wire 45 disconnected at the SCR terminal board, check for control volts positive at the SCR terminal board (wire 41) to negative (wire 13A). If there are zero volts at this point, check F or R normally open interlocks and the 1A coil for continuity. • (See NOTE 1) With F or R picked up and wire 45 disconnected at the SCR terminal board, check for positive control volts at the 1 REC heat sink (wire 25) to negative (wire 13A). If there are zero volts at this point, check: The FUB, F or R power tips, and the continuity of wiring from the battery positive to the 1 REC heat sink.

NOTE 1: The drive wheels should be off the floor.

1C (Continued)	• (See NOTE 1) With F or R picked up and wire 45 disconnected from the SCR terminal board, measure approximately 6 volts from (wire 29) to negative (wire 13A) with the accelerator pot near creep speed. Volts will drop to zero as the accelerator is moved toward full speed. If the readings are not correct, first place a jumper wire between wires 29 and 29A, which bypasses the thermal protector. Depress the accelerator and check for the above voltage. If the voltage readings are correct, replace the thermal protector. If the above tests will produce no voltage change, place a jumper between wires 29A and 13A. This bypasses the accelerator and the truck should now run at top SCR speed. If top speed is obtained, check the accelerator pot per Table 4J. If the motor fails to operate, check the card per Table 4A. • Check 1 REC for open circuit or open gate (see 4I). • Check the card (see 4A).
1D. The contactors close, but very little power and a high-pitch SCR hum are evidenced.	• Check 2 REC for a shorted condition in the conducting direction (see 4I).
1E. The contactors close. Very little or no power is present with a low SCR hum, even when the accelerator is in the top SCR position.	• (See NOTE 1) Disconnect wire 5 from the sensor. Reapply power, and if the control operates normally, check the settings of the current limit, and plugging.
1F. The contactors close. There is very little power and a normal SCR hum.	• Check 3 REC for open condition (see 4H). If 3 REC is found to be open, check 1, 2, and 5 REC for proper operation. • Check 4 REC for short (see 4H). *NOTE 1: The drive wheels should be off the floor.*

Table 2

Failures Which Cause Full Motor Torque With SCR Control

SYMPTOMS	WHAT TO DO
2A. The contactors close. Full SCR speed is immediately attainable with an audible hum.	• Check the potentiometer for proper resistance (see 4J). • Check for grounds in wires 29 and 29A, or a shorted accelerator pot.
2B. The contactors close. Full speed is immediately attainable with no audible hum. *	• Check for welded power tips on the 1A contactor. • Check the timer section of the card (see 4Ab).
2C. The contacts close. Full speed is immediately attainable with no audible hum. * The capacitor is not charged.	• Check for an open gate circuit to 5 REC (see 4I and 4Ba). • Check 5 REC for shorted condition (see 4I). If 5 REC shorted, also check 4Bc. • Check the continuity of the wiring from 1C to 5 REC and from 5 REC through T3, T4 to T1, and 1 REC wire 25. • Check capacitor 1C (see 4E). • Check 1 REC for short (see 4I).
2D. The contactors close. Full speed is immediately attainable with no audible hum. * The capacitor is charged.	• Check for an open 2 REC (see 4I). • Check for an open gate in 2 REC (see 4I). • Check for an open gate circuit to 2 REC (see 4Bb).

* If the truck is equipped with a fault detector and it fails to shut down the control on the above faults, check the fault detector per GEK-7940.

Table 3

Misoperation of Special Features

SYMPTOMS	WHAT TO DO
3A. Failure of the 1A contactor to operate.	• (See NOTE 1) With all direction switches closed, jumper negative to the SCR terminal board (wire 41). 1A should pick up immediately. This checks the 1A coil. • (See NOTE 1) Move the negative jumper to the SCR terminal board (wire 45). 1A should pick up after approximately a 1 second delay. This checks the timer section of Card 1. If 1A does not operate check the Timer Section of Card 1 (see 4Ab).
3B. Failures in the FW circuit.	• See GEK-7942.
3C. Severe reversal.	• Check the settings of the plugging trimpot on Card 1 (see 6Ad). • Check 4 REC (see 4H). • Check the current limit setting (see note in 6Ac). • Check continuity of wires 1 and 5. • Check FUA (if used).
3D. Severe reversal at the end of the plug.	• Check the setting of the plug trimpot. Setting this trimpot too far counterclockwise tends to make plug more severe at the end of the plug. (See SEC.)

NOTE 1: The drive wheels should be off the floor.

Table 4

Checking Components

WARNING: Before touching electrical components, disconnect the battery and discharge capacitor 1C.

4A CARD 1 (See Table 6 for tuneup of Card 1).

Remove the card from the panel by loosening the two screws at the bottom of the box, pull the box straight up to disengage from the receptacle. Connections can be made to the card pins with insulated clips.

a) Oscillator Section

Connect a 1 MFD capacitor and a volt-ohmmeter positive to 37, negative to 13. Connect a 4.7K resistor from 29 to 13. Using a 36-volt test battery, connect the battery positive through a 220-ohm, 2-watt resistor to terminals 41 and 25. Connect the battery negative to terminal 13. The meter should read less than 10 volts. Adjust the creep speed trimpot until the voltmeter reads .5 volts. Remove the capacitor. The voltmeter should now read less than .1 volt.

b) 1A Timer

Connect the volt-ohmmeter positive to 41, the negative to 45, and set the scale to 50-volts d-c. Using a 36-volt test battery, connect the battery positive through a 220-ohm, 2-watt resistor to terminal 41. Connect the battery negative through a normally open switch to terminal 45. Close the switch and observe the battery voltage on the VOM. After approximately 1 second, voltage should drop to 0, indicating timer action.

4B FILTER AND FIRING BLOCK

The filter and firing circuits for the transformer, 2 REC and 5 REC, are located in a separate filter and firing block. These circuits can be checked with a volt-ohmmeter.

a) 5 REC Firing Circuit:

Set the VOM on the RX100 scale. Disconnect the lead to terminal 7. Connect the VOM positive lead to terminal 8, and the negative lead to terminal 7. The circuit should now read 1700 to 2100 ohms. Reverse the leads and read infinity.

b) 2 REC Firing Circuit:

Set the VOM on the RX100 scale. Disconnect the lead to terminal 5. Connect the VOM positive lead to terminal 6, and the negative lead to terminal 5. The circuit should read 1170 to 1430 ohms. Reverse the leads and read infinity.

c) Transformer Filter:

Set the VOM on the RX100 scale. Disconnect the lead to terminal 4. Connect the VOM positive lead to terminal 6, and the negative lead to terminal 4. The circuit should read 1700 to 2100 ohms. Reverse the leads and read infinity.

4C FIELD WEAKENING CARD (if used)

See GEK-7942.

4D FAULT DETECTOR (if used)

See GEK-7940.

4E CAPACITOR 1C

Disconnect the battery and discharge the capacitor. Remove Card 1. Measure ohms through the capacitor using the RX10, 000 scale. The meter should read zero ohms and then swing to above 100, 000 ohms. Replace the capacitor if above reading is not obtained.

4F CONTACTORS F, R, 1A, FW AND P

a) 100-ampere contactors (See GEH-3101)

b) 200-ampere contactors (see GEH-3074)

NOTE: The control is arranged so that F and R do not normally break current. Contactor 1A drops out ahead of F or R.

4G CONTACTOR COIL AND ACCESSORY FILTER (7, 8, 9, and 12 REC)

On some magnetic panels, the contactor coils will either be varnished-tape-wound or encapsulated in green epoxy. For the varnished-tape-wound type, a separate filter is required and will be mounted adjacent to the coil. The new green epoxy encapsulated coil contains the necessary filtering and is not visible from the exterior of the device.

a) Separate Filter (Typical Cat. No. 148B6203G14)

These are varistors and should be checked as follows: *DISCONNECT THE BATTERY AND DISCHARGE CAPACITOR 1C.* Disconnect the leads to the filter block. Connect a 36-volt d-c test battery in series with the varistor, and a volt-ohmmeter set on the 1ma. scale as shown in Figure 7. If the varistor is good, there will be a noticeable deflection of the meter needle when the leads are touched to the filter block terminals. If no deflection is obtained, replace the filter block.

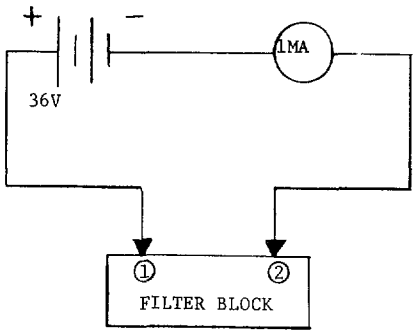


Figure 7

b) Integral Coil Filter

When this filter fails, it will be evident by a severe cracking of the coils in the vicinity of the coil terminals.

4H RECTIFIERS

When checking diodes, *DISCONNECT THE BATTERY AND DISCHARGE CAPACITOR 1C* to prevent burning out the ohmmeter. When reassembling the rectifiers, refer to Table 5.

3 and 4 REC: Disconnect the pigtail. 3 and 4 REC are diodes with about 7 to 12 ohms in the conducting direction (+ ▶ −) measured on the RX1 scale, and infinite resistance in the nonconducting direction (− ▶ +) measured on the RX10, 000 scale.

15 and 16 REC: Disconnect one lead. Check in the same manner as 3 and 4 REC above.

4I SCR'S (1 REC, 2 REC, 5 REC)

These are silicon control rectifiers. Before checking, *DISCONNECT BATTERY AND DISCHARGE CAPACITOR 1C*. Before checking 1 REC, remove Card 1 to open the gate and disconnect the pigtail. Before checking 2 REC disconnect the gate lead by opening the lead to terminal 5 of the filter and firing block, and disconnect the pigtail. Before checking 5 REC disconnect the gate lead by opening the lead to terminal 7, and disconnect the power lead to the terminal on top of 5 REC. Continue checking as indicated below.

To check an SCR, it is necessary to have a 3-volt battery and a 3-volt lamp. (A test flashlight such as a BRIGHT STAR No. 1618CT circuit continuity tester is excellent for this test.)

Connect the positive lead to the stud (1), and connect the negative lead to the pigtail (3), as shown in Figure 8.

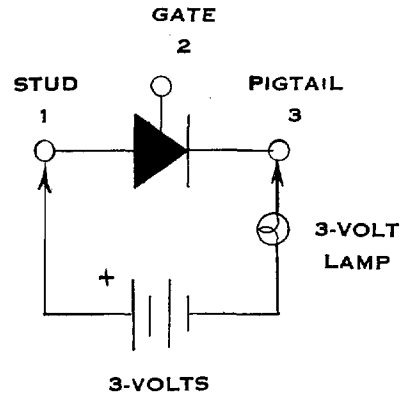


Figure 8

a) The lamp should not light. If the lamp does light, the SCR is shorted and must be replaced.

b) If check (a) was satisfactory, test the SCR for its ability to be turned on by the gate. Touch gate (point 2) to point 1. If gate is operative, the lamp should come on and must remain on when the gate is removed.

c) If lamp cannot be lit under step (b) the SCR is open and must be replaced.

When reassembling SCR's, refer to Table 5.

4J ACCELERATOR POTENTIOMETER

To check the operation of the potentiometer, disconnect the battery and disconnect wire 29A from the thermal protector or the SCR terminal board. Connect the VOM from wire 29A to negative (13A) with the scale set to RX 100. With the accelerator in creep-speed position, the ohm reading should be 3,500 to 6,600 ohms; with the accelerator in top-speed position, the reading should be 200 ohms or less. If these readings are not obtained, loosen the pinion gear clamp and adjust the rotation of the plot shaft relative to the accelerator shaft, or replace the shaft.

With the wires disconnected as described above, check for a resistance of 1 megohm or higher from the pot wires to the truck frame.

For additional information, refer to GEK-8073.

4K THERMAL PROTECTOR (TP)

Remove both connections from the TP and, with a VOM, read approximately 50 ohms from terminal to terminal, if the heat sink is at room temperature. Set the VOM to the highest ohm scale and check the pins to the heat sink. This reading should be infinity.

4L FILTER BLOCK (HF)

To check, disconnect all wires from the filter block. With the VOM on the RX 10, 000 scale, touch the leads to the filter terminals in order to charge the filter. After a few seconds, reverse the meter leads and touch the filter terminals. The VOM needle will deflect and return to infinity. If this capacitor action is not observed, replace the filter block.

Table 5

Replacement of Semiconductors

When replacing semiconductors such as 1, 2, 3, 4 and 5 REC; it is not necessary to torque these devices to a specific value. However, the device should be screwed into the heat sink and tightened to a snug fit.

The use of a heat transfer grease (such as GE Versilube G-350-M or equivalent) is recommended.

Locking devices must be properly adjusted to prevent the semiconductor from becoming loose.

Table 6

Tuneup for New or Mistuned Card 1

IMPORTANT NOTES

1. Panels are factory-adjusted for a particular motor and truck and should not need adjustment when used with this motor and truck. However, checks and or touchup adjustments may be made per Table 6A, because these adjustments are so designed that they do not interact when near their proper setting.

2. The TOP SCR SPEED setting is a factory made and sealed setting! Under normal conditions, this setting should not be touched.

   If setting is required, the complete tuneup procedure, Table 6B, must be followed.

3. If the panels are used to control motors or trucks for which they were not factory adjusted, the settings may be out of optimum adjustment to the extent that they do interact, and the complete tuneup procedure, Table 6B, must then be followed.

4. All adjustments are such that CW rotation increases the function being adjusted.

5. Connect the shunt, the millivoltmeter and the voltmeter to measure battery current and motor volts. Connect the shunt and millivoltmeter between battery negative and 1 REC (or between truck receptacle and battery plug). Connect voltmeter between battery positive and T2 on the SCR panel. Remove wire 29 from the thermal protector and connect a jumper from wire 29 to negative (pigtail of 1 REC).

   NOTE: Be sure to insulate or wrap the jumper connection to wire 29 to prevent accidental contact of this point to the truck frame. If this point touches the truck frame, it will damage the control card.

   Jack up the truck so that the drive wheels are free to rotate.

   If a brake interlock is used, jumper it out so that power and brakes can be applied at the same time.

6. Equipment required:

   — 50-millivolt d-c shunt*

   — 50-millivolt d-c meter (d'Arsonval movement)

   — 50-volt d-c meter (d'Arsonval movement) (250 volt scale needed for 72V)

   — Battery with an equal or greater ampere-hour capacity than used on the truck, charged to 1200 or higher specific gravity.

*Typical Shunt Ratings	SCR Model
300 or 400 Amp	201
400 or 500 Amp	301

NOTE: The shunt rating must be greater than the current to be measured. Best results are obtained when the reading is between half and full scale on meter. If a shunt of too high a rating is used (i.e., a 500 amp rating to read 100 amps), it will be hard to read and the accuracy of the reading will be poor.

7. Check that the ohms in the accelerator potentiometer are less than 200 ohms in the top SCR range (see Table 4I). Refer to Figure 3 for potentiometer locations.

Table 6A

Checking of Card Settings

Checks and/or minor touchup adjustments can be made without following complete tuneups as given in Table 6B.

a) CREEP SPEED

With the truck on the ground check the creep speed of the truck as either the F or R contactor operates.

b) TOP SCR SPEED

Refer to Note 5.

Check TOP SCR SPEED by first moving the accelerator until the F or R contactor operates. Do not move the accelerator to the point where 1A picks up.

Apply the brakes until the battery current reads the value given in Table 6Bd, and then read motor voltage to see if it falls within the values given in Table 6Bd.

Disconnect the shunt, voltmeter, and jumper to wire 29, and reconnect wire 29 on the thermal protector.

c) CURRENT LIMIT

Refer to Note 5.

Check the CURRENT LIMIT by first moving the accelerator until either the F or R contactor operates. Do not move the accelerator to the point where 1A picks up.

Apply the brakes until the wheels come to a STANDSTILL (the wheels must not be turning) and read the current to see if it falls below the maximum rating given in Table 6Bd and within the rating specified by the truck manufacturer.

NOTE: DO NOT STALL the motor for more than 30 seconds at a time. Allow time for motor cooling between stalls. Do not operate the motor at high speeds or plug the motor with wheels jacked up.

NOTE: When setting the current limit trimpot DO NOT set it further clockwise than where the maximum current or current limit setting is obtained. SETTING THE CURRENT TRIMPOT TOO FAR CLOCKWISE CAN CREATE A SEVERE PLUG.

Disconnect the shunt, voltmeter, and jumper to wire 29, and reconnect wire 29 on the thermal protector.

d) STATIC PLUGGING

With the truck on the ground, plug the truck from top speed. If the stopping distance is not as desired, adjust the plugging trimpot.

e) 1A TIMER

With the truck on the ground check the operation for full acceleration. If the 1A contactor picks up too early, or if the truck is sluggish, adjust the trimpot to obtain desired results.

Table 6B

Tuneup Procedure

COMPLETE ALL STEPS a through e.

a) Adjust CREEP SPEED as desired.

b) TOP SCR SPEED

Refer to Note 5.

Check TOP SCR SPEED by first moving the accelerator until the F or R contactor operates. Do not move the accelerator to the point where 1A picks up.

Apply the brakes until the battery current is about 70 to 80 percent of the loaded level amperes with the motor directly across the battery and the motor fields connected the same as in the SCR control.

Typical Battery Current	SCR Model
75 - 80 Amps	201
120 - 130 Amps	301

Adjust the TOP SCR SPEED trimpot until the voltmeter reads motor voltage as indicated below:

		CURRENT LIMIT AMPS (MAXIMUM)	
Battery Volts	Motor Volts	C185 Model 201	C290 Model 301
12	7 - 9	200	300
18	13 - 16	200	300
24	17 - 22	200	300
36	25 - 32	200	300
48	34 - 43	185	300
72	50 - 65	150	300

c) CURRENT LIMIT

Turn the current limit trimpot fully counterclockwise.

Depress the accelerator until either the F or R contactor, (but not the 1A contactor) operates.

Apply the brakes until the wheels come to a standstill and remain at a standstill.

Slowly turn the CURRENT LIMIT TRIMPOT, in a clockwise direction, until the current reaches a value as shown in the above table, or as specified by the truck manufacturer.

NOTE: The current limit values as given in the table above represent the typical maximum values that we suggest for each control when operated on a given voltage and these values must not be exceeded.

Since these controls are used on a variety of types and sizes of trucks for various applications, it is common for the truck manufacturer to set the current limit at some value below this maximum. For this reason it is recommended that you obtain the actual current limit setting for your particular truck from the truck manufacturer.

NOTE: Do not stall the motor for more than 30 seconds at a time. Allow time for motor cooling between stalls. Do not operate the motor at high speeds or plug the motor with the wheels jacked up.

After setting to the correct value, disconnect the meter and reconnect wire 29 to the thermal protector.

d) STATIC PLUGGING

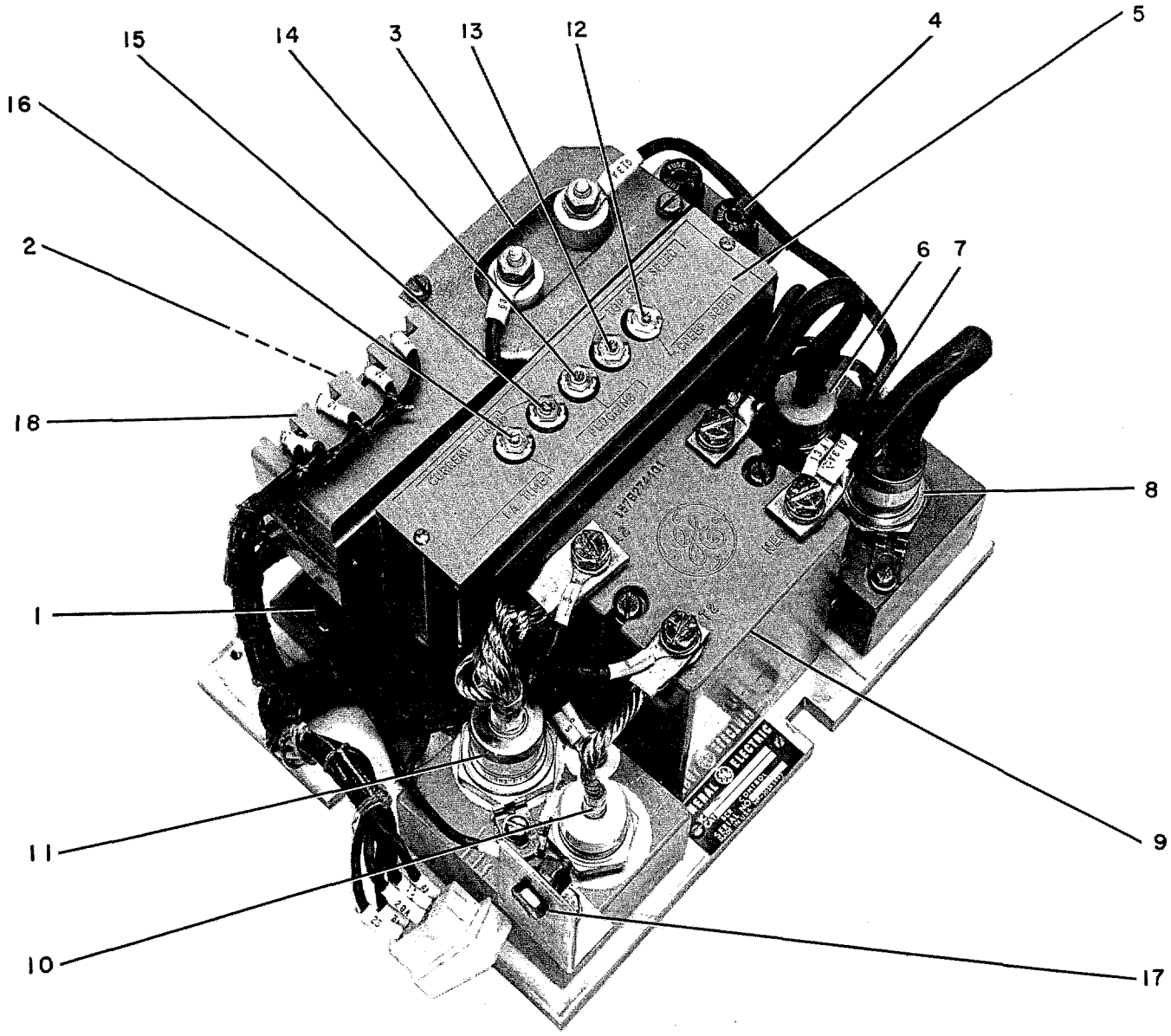
To adjust the static plugging, the truck should be in its normal running condition, and on the ground.

Turn the PLUGGING TRIMPOT fully counterclockwise, then rotate it about 30 degrees clockwise. Run the truck to check the stopping distance. Turning the trimpot clockwise will shorten the stopping distance; a counterclockwise rotation will lengthen the stopping distance.

NOTE: The plugging trimpot should never be left in the extreme counterclockwise direction as it may affect the smoothness of truck re-acceleration.

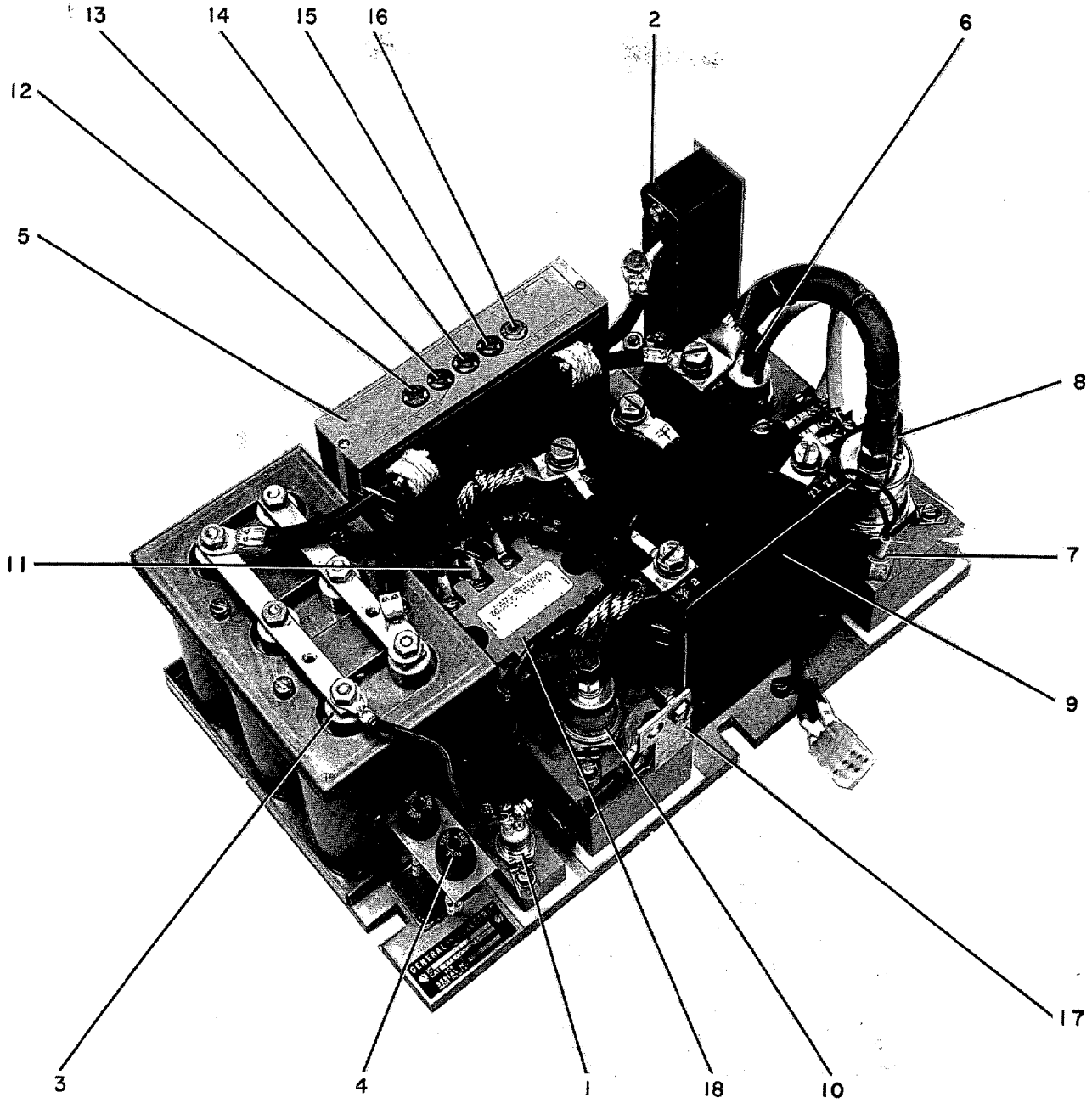
TYPICAL PHYSICAL ARRANGEMENT AND IDENTIFICATION OF COMPONENTS

(Refer to wiring diagram furnished with truck for precise arrangement of components.)



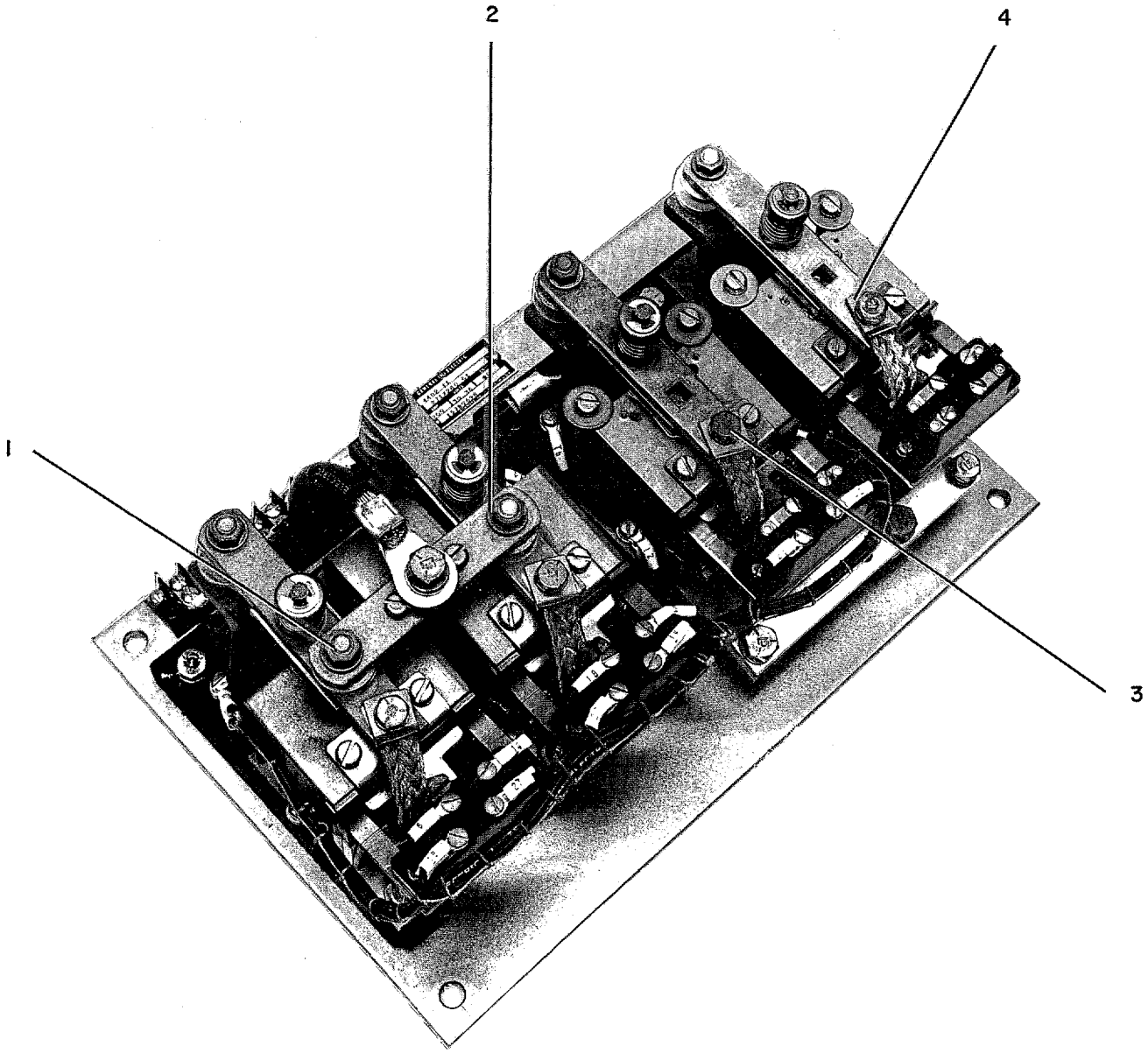
(1) BLOCKING SCR (5 REC)
(2) CURRENT-LIMITING REACTOR (1X)
(3) COMMUTATING CAPACITOR (1C)
(4) CONTROL CIRCUIT FUSES
(5) OSCILLATOR CARD (CARD 1)
(6) TURN-OFF SCR (2 REC)
(7) THERMAL PROTECTOR
(8) MAIN SCR (1 REC)
(9) PULSE TRANSFORMER (1T)
(10) FLY-BACK DIODE (3 REC)
(11) PLUGGING DIODE (4 REC)
(12) CREEP-SPEED ADJUSTMENT
(13) TOP-SPEED ADJUSTMENT
(14) STATIC-PLUGGING ADJUSTMENT
(15) CURRENT-LIMIT ADJUSTMENT
(16) 1A TIMER ADJUSTMENT
(17) SENSOR
(18) FILTER BLOCK

Figure 9. Typical SCR Static Panel
Model 201



(1) BLOCKING SCR (5 REC)
(2) CURRENT-LIMITING REACTOR (1X)
(3) COMMUTATING CAPACITOR (1C)
(4) CONTROL CIRCUIT FUSES
(5) OSCILLATOR CARD (CARD 1)
(6) TURN-OFF SCR (2 REC)
(7) THERMAL PROTECTOR
(8) MAIN SCR (1 REC)
(9) PULSE TRANSFORMER (1T)
(10) FLY-BACK DIODE (3 REC)
(11) PLUGGING DIODE (4 REC)
(12) CREEP-SPEED ADJUSTMENT
(13) TOP-SPEED ADJUSTMENT
(14) STATIC-PLUGGING ADJUSTMENT
(15) CURRENT-LIMIT ADJUSTMENT
(16) 1A TIMER ADJUSTMENT
(17) SENSOR
(18) FILTER BLOCK

Figure 10. Typical SCR Static Panel Model 301



(1) FORWARD CONTACTOR (F)
(2) REVERSE CONTACTOR (R)
(3) BY-PASS CONTACTOR (1A)
(4) PUMP CONTACTOR (P)

Figure 11. Typical Magnetic Panel

INDUSTRY CONTROL DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

8-69 (5M)